Jiayun Meng

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Education

August 2022 - **Ph.D. in Mathematics**, *The University of Texas at Austin*, USA.

- Advisor: Alexis Vasseur

September 2018 - June 2022 **B.S. in Mathematics**, *University of California, Los Angeles*, USA.

- Departmental honors

Research interests

Partial differential equations, compressible fluid dynamics, hyperbolic conservation laws

Publications

Published papers and Preprints

[1] Michael Hill, Jiayun Meng, and Nan Li. Counting compatible indexing system for C_{p^n}. *Orbita Mathematicae 1 (2024), 37-61.*

[2] Jiayun Meng. Time-asymptotic stability of composite weak planar waves for a general n×n multi-D viscous system. *arXiv preprint arXiv:2501.14188, 2025. Submitted.*

[3] Jiayun Meng and Alexis Vasseur. Weak-BV stability for the impermeable wall problem of the isentropic Euler and its inviscid limit from Navier-Stokes. *Discrete and Continuous Dynamical Systems*, 59(0):29-44, 2027.

[4] Moon-Jin Kang, Jiayun Meng, HyeonSeop Oh, and Alexis Vasseur. Weak-BV stability of inflow and outflow problems for the isentropic Euler system. *arXiv preprint arXiv:2610.03236, 2026. Submitted.*

[5] Jiayun Meng and Alexis Vasseur. Weak-BV stability and inviscid limits for barotropic flows with outflow boundary conditions. *arXiv preprint arXiv:2610.07574, 2026. Submitted.*

In preparation

[1] Young-Sam Kwon, Jiayun Meng, and Alexis Vasseur. a-contraction applied to the time-asymptotic stability of weak shocks for a general 1-d viscous system. *In preparation.*

Invited Talks

- Aug. 2026 Conference on Analysis, Modeling and Numerics of Waves, *Delft University of Technology*
- July 2026 2026 SIAM Annual Meeting, *Huntington Convention Center of Cleveland*
- June 2026 The 20th International Conference on Hyperbolic Problems: Theory, Numerics and Applications, *University of Stuttgart*
- Nov. 2024 Workshop on Stability Analysis for Nonlinear PDEs and Multiscale Applications, *University of Wisconsin-Madison*
- Oct. 2024 SIAM Conference Texas-Louisiana Section, *Baylor University*
- Aug. 2024 PDE and Applications seminar, *Chinese Academy of Sciences*

- July 2024 The 19th International Conference on Hyperbolic Problems: Theory, Numerics and Applications, *Shanghai Jiao Tong University*

Other presentations

Posters

- Nov. 2024 Paris-Seoul Joint Workshop on Fluid Dynamics 2025, École normale supérieure

International visits

- May 11-15 2026 *Institut des Hautes Études Scientifiques*
- Aug. 11-15 2025 *Korea Advanced Institute of Science & Technology (KAIST)*, hosted by Professor Moon-Jin Kang

Journal referee service

- 2026 *Communications in Mathematical Sciences*

Honors and Awards

- Summer 2025 Summer Excellence Fellowship, UT Austin
- Summer 2025 Professional Development Award, UT Austin

Teaching Experience

Teaching Assistant

- Spring 2026 M365C Real analysis I
- Fall 2025 M381C Real Analysis (Graduate)
- Spring 2025 M383D Methods of Applied Mathematics II (Graduate)
- Spring 2024 M383D Methods of Applied Mathematics II (Graduate)
- Fall 2023 M427J Differential Equations with Linear Algebra
- Spring 2023 M408D Sequences, Series, and Multivariable Calculus
- Fall 2022 M408C Differential and Integral Calculus

Laboratory Assistant

- Fall 2019 - Winter 2020 UCLA Olga Radko Endowed Math Circle